The Postulates of Quantum Mechanics (Cohen–Tannoudji, Ch. III) Section B. The physical interpretation of the postulates

Beatriz Gonzalez del Rio

Outline

- A. Statement of the postulates
- B. The physical interpretation of the postulates concerning observables and their measurements
- C. The physical implications of the Schrödinger equation

Probability interpretation

For the position observable X, the wave function is

$$\psi(x) = \langle x|\psi\rangle\,.$$

Probability interpretation

For the position observable X, the wave function is

$$\psi(x) = \langle x|\psi\rangle\,.$$

The probability of finding the particle between x and $x + dx$ is:

$$d\mathcal{P}(x) = |\psi(x)|^2\, dx.$$

Probability interpretation

For the position observable X, the wave function is

$$\psi(x) = \langle x|\psi\rangle\,.$$

The probability of finding the particle between x and $x + dx$ is:

$$d\mathcal{P}(x) = |\psi(x)|^2\,dx.$$

This is the probabilistic interpretation of the wave function.

Momentum representation

The eigenvectors $|p\rangle$ of the momentum operator correspond to plane waves:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\, e^{ipx/\hbar}.$$

Momentum representation

The eigenvectors $|p\rangle$ of the momentum operator correspond to plane waves:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\, e^{ipx/\hbar}.$$

The momentum-space wave function is:

$$\tilde{\psi}(p) = \langle p|\psi\rangle\,.$$

Momentum representation

The eigenvectors $|p\rangle$ of the momentum operator correspond to plane waves:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\, e^{ipx/\hbar}.$$

The momentum-space wave function is:

$$\tilde{\psi}(p) = \langle p|\psi\rangle\,.$$

The probability of finding a momentum between p and $p + dp$ is:

$$d\mathcal{P}(p) = |\tilde{\psi}(p)|^2\, dp.$$

Quantization of physical quantities

The third postulate explains why some observables have discrete spectra.

Quantization of physical quantities

The third postulate explains why some observables have discrete spectra.

However, not all physical quantities are quantized: some observables have continuous spectra.

Quantization of physical quantities

The third postulate explains why some observables have discrete spectra.

However, not all physical quantities are quantized: some observables have continuous spectra.

Discrete values emerge from solving the eigenvalue equation of the corresponding operator.

The measurement process

Measurement involves an interaction between the system and the measuring device.

The measurement process

Measurement involves an interaction between the system and the measuring device.

The fourth and fifth postulates introduce an intrinsic nondeterminism in the outcome of measurements.

The measurement process

Measurement involves an interaction between the system and the measuring device.

The fourth and fifth postulates introduce an intrinsic nondeterminism in the outcome of measurements.

Only probabilities for measurement outcomes can be predicted.

Mean value of an observable

The mean value of an observable A in the state $|\psi\rangle$ is defined as the average result obtained from a large number of identical measurements.

Mean value of an observable

The mean value of an observable A in the state $|\psi\rangle$ is defined as the average result obtained from a large number of identical measurements.

Two cases must be distinguished:

- observables with a **discrete spectrum**,
- observables with a **continuous spectrum**.

Mean value of an observable

The mean value of an observable A in the state $|\psi\rangle$ is defined as the average result obtained from a large number of identical measurements.

Two cases must be distinguished:

- observables with a **discrete spectrum**,
- observables with a **continuous spectrum**.

We now derive the expression of $\langle A \rangle_\psi$ in both cases.

Mean value: discrete spectrum

Let A have a discrete spectrum $\{a_n\}$, with eigenvectors $|u_n\rangle$:

$$A|u_n\rangle = a_n|u_n\rangle .$$

Mean value: discrete spectrum

Let A have a discrete spectrum $\{a_n\}$, with eigenvectors $|u_n\rangle$:

$$A\,|u_n\rangle = a_n\,|u_n\rangle\,.$$

The probability of obtaining a_n in the state $|\psi\rangle$ is:

$$\mathcal{P}(a_n) = |\,\langle u_n|\psi\rangle\,|^2.$$

Mean value: discrete spectrum

Let A have a discrete spectrum $\{a_n\}$, with eigenvectors $|u_n\rangle$:

$$A\,|u_n\rangle = a_n\,|u_n\rangle\,.$$

The probability of obtaining a_n in the state $|\psi\rangle$ is:

$$\mathcal{P}(a_n) = |\,\langle u_n|\psi\rangle\,|^2.$$

After N measurements, the mean value is:

$$\langle A\rangle_\psi = \sum_n a_n\,\mathcal{P}(a_n).$$

Mean value: discrete spectrum

Substituting the probability:

$$\langle A \rangle_\psi = \sum_n a_n \, \langle \psi | \, u_n \rangle \langle u_n \, | \psi \rangle \, .$$

Mean value: discrete spectrum

Substituting the probability:

$$\langle A \rangle_\psi = \sum_n a_n \, \langle \psi | \, u_n \rangle \langle u_n \, | \psi \rangle \, .$$

Using the completeness relation:

$$\sum_n |u_n\rangle \, \langle u_n| = \mathbb{I},$$

we obtain:

$$\boxed{\langle A \rangle_\psi = \langle \psi | \, A \, | \psi \rangle}.$$

Mean value: continuous spectrum

Let A have a continuous spectrum α, with eigenstates $|v_\alpha\rangle$:

$$A\,|v_\alpha\rangle = \alpha\,|v_\alpha\rangle\,.$$

Mean value: continuous spectrum

Let A have a continuous spectrum α, with eigenstates $|v_\alpha\rangle$:

$$A\,|v_\alpha\rangle = \alpha\,|v_\alpha\rangle\,.$$

The probability of obtaining a result between α and $\alpha + d\alpha$ is:

$$d\mathcal{P}(\alpha) = |\,\langle v_\alpha|\psi\rangle\,|^2\,d\alpha.$$

Mean value: continuous spectrum

Let A have a continuous spectrum α, with eigenstates $|v_\alpha\rangle$:

$$A\,|v_\alpha\rangle = \alpha\,|v_\alpha\rangle\,.$$

The probability of obtaining a result between α and $\alpha + d\alpha$ is:

$$d\mathcal{P}(\alpha) = |\,\langle v_\alpha|\psi\rangle\,|^2\,d\alpha.$$

The mean value is therefore:

$$\langle A\rangle_\psi = \int \alpha\,d\mathcal{P}(\alpha).$$

Mean value: continuous spectrum

Substituting the probability density:

$$\langle A \rangle_\psi = \int \alpha \ \langle \psi | \, v_\alpha \rangle \langle v_\alpha \, | \psi \rangle \ d\alpha.$$

Mean value: continuous spectrum

Substituting the probability density:

$$\langle A \rangle_\psi = \int \alpha \ \langle \psi | \, v_\alpha \rangle \langle v_\alpha \, | \psi \rangle \ d\alpha.$$

Using the closure relation:

$$\int d\alpha \ |v_\alpha\rangle \, \langle v_\alpha| = \mathbb{I},$$

we again obtain:

$$\boxed{\langle A \rangle_\psi = \langle \psi | \, A \, | \psi \rangle}.$$

Important remarks on mean values

(i) Mean value vs. time average

Important remarks on mean values

(i) Mean value vs. time average

The mean value $\langle A \rangle_\psi$ must **not** be confused with a time average taken over the evolution of a single system.

Important remarks on mean values

(i) Mean value vs. time average

The mean value $\langle A \rangle_\psi$ must **not** be confused with a time average taken over the evolution of a single system.

It corresponds to an average over a large number of **identical measurements performed on identically prepared systems**.

Important remarks on mean values

(i) Mean value vs. time average

The mean value $\langle A \rangle_\psi$ must **not** be confused with a time average taken over the evolution of a single system.

It corresponds to an average over a large number of **identical measurements performed on identically prepared systems**.

(ii) Non-normalized states

Important remarks on mean values

(i) Mean value vs. time average

The mean value $\langle A \rangle_\psi$ must **not** be confused with a time average taken over the evolution of a single system.

It corresponds to an average over a large number of **identical measurements performed on identically prepared systems**.

(ii) Non-normalized states

If the state $|\psi\rangle$ is not normalized, the mean value is given by:

$$\boxed{\langle A \rangle_\psi = \frac{\langle \psi | A | \psi \rangle}{\langle \psi | \psi \rangle}}$$

Expectation values in a representation

In the position representation:

$$\langle X \rangle_\psi = \langle \psi | \, X \, | \psi \rangle = \int d^3 r \, \langle \psi | r \rangle \, \langle r | \, X \, | \psi \rangle = \int d^3 r \, \psi^*(r) \, x \, \psi(r).$$

Expectation values in a representation

In the position representation:

$$\langle X \rangle_\psi = \langle \psi | \, X \, | \psi \rangle = \int d^3 r \, \langle \psi | r \rangle \, \langle r | \, X \, | \psi \rangle = \int d^3 r \, \psi^*(r) \, x \, \psi(r).$$

For the momentum operator:

$$\langle P_x \rangle_\psi = \int d^3 r \, \psi^*(r) \left(-i\hbar \frac{\partial}{\partial x} \right) \psi(r).$$

Expectation values in a representation

In the position representation:

$$\langle X \rangle_\psi = \langle \psi | X | \psi \rangle = \int d^3 r \, \langle \psi | r \rangle \, \langle r | X | \psi \rangle = \int d^3 r \, \psi^*(r) \, x \, \psi(r).$$

For the momentum operator:

$$\langle P_x \rangle_\psi = \int d^3 r \, \psi^*(r) \left(-i\hbar \frac{\partial}{\partial x} \right) \psi(r).$$

Expectation values are computed using the operator representation in the chosen basis.

Dispersion of measurement results

The mean value $\langle A \rangle_\psi$ gives the order of magnitude of the values obtained when measuring the observable A.

Dispersion of measurement results

The mean value $\langle A \rangle_\psi$ gives the order of magnitude of the values obtained when measuring the observable A.

However, it does **not** provide information about the dispersion of the measurement results around $\langle A \rangle_\psi$.

Dispersion of measurement results

The mean value $\langle A \rangle_\psi$ gives the order of magnitude of the values obtained when measuring the observable A.

However, it does **not** provide information about the dispersion of the measurement results around $\langle A \rangle_\psi$.

For a continuous spectrum, the results are distributed according to a probability density $\rho(\alpha)$, typically concentrated around $\langle A \rangle_\psi$.

Dispersion of measurement results

The mean value $\langle A \rangle_\psi$ gives the order of magnitude of the values obtained when measuring the observable A.

However, it does **not** provide information about the dispersion of the measurement results around $\langle A \rangle_\psi$.

For a continuous spectrum, the results are distributed according to a probability density $\rho(\alpha)$, typically concentrated around $\langle A \rangle_\psi$.

We therefore need a quantity that characterizes the **width of the distribution**.

Why simple deviations do not work

A natural idea is to consider, for each measurement, the deviation from the mean:

$$A - \langle A \rangle.$$

Why simple deviations do not work

A natural idea is to consider, for each measurement, the deviation from the mean:

$$A - \langle A \rangle.$$

Taking the average of these deviations gives:

$$\langle A - \langle A \rangle \rangle = \langle A \rangle - \langle A \rangle = 0.$$

Why simple deviations do not work

A natural idea is to consider, for each measurement, the deviation from the mean:

$$A - \langle A \rangle.$$

Taking the average of these deviations gives:

$$\langle A - \langle A \rangle \rangle = \langle A \rangle - \langle A \rangle = 0.$$

Positive and negative deviations compensate exactly.

Why simple deviations do not work

A natural idea is to consider, for each measurement, the deviation from the mean:

$$A - \langle A \rangle.$$

Taking the average of these deviations gives:

$$\langle A - \langle A \rangle \rangle = \langle A \rangle - \langle A \rangle = 0.$$

Positive and negative deviations compensate exactly.
To avoid this cancellation, we must consider the **squares of the deviations**.

Root-mean-square deviation

We define the square of the dispersion as the mean of the squared deviations:

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle.$$

Root-mean-square deviation

We define the square of the dispersion as the mean of the squared deviations:

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle.$$

The quantity ΔA is called the **root-mean-square deviation** of A.

Root-mean-square deviation

We define the square of the dispersion as the mean of the squared deviations:

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle.$$

The quantity ΔA is called the **root-mean-square deviation** of A. By definition:

$$\boxed{\Delta A = \sqrt{\langle (A - \langle A \rangle)^2 \rangle}}$$

Root-mean-square deviation

We define the square of the dispersion as the mean of the squared deviations:

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle.$$

The quantity ΔA is called the **root-mean-square deviation** of A.
By definition:

$$\boxed{\Delta A = \sqrt{\langle (A - \langle A \rangle)^2 \rangle}}$$

This quantity measures how strongly the measurement results are spread around the mean value.

Equivalent expression for ΔA

Expanding the square:

$$(A - \langle A \rangle)^2 = A^2 - 2\langle A \rangle A + \langle A \rangle^2.$$

Equivalent expression for ΔA

Expanding the square:

$$(A - \langle A \rangle)^2 = A^2 - 2\langle A \rangle A + \langle A \rangle^2.$$

Taking the mean value:

$$\langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2.$$

Equivalent expression for ΔA

Expanding the square:

$$(A - \langle A \rangle)^2 = A^2 - 2\langle A \rangle A + \langle A \rangle^2.$$

Taking the mean value:

$$\langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2.$$

Therefore, the root-mean-square deviation can be written as:

$$\boxed{\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}}$$

Continuous spectra and uncertainty relations

For a continuous spectrum with probability density $\rho(\alpha)$, the dispersion is given by:

$$(\Delta A)^2 = \int_{-\infty}^{+\infty} (\alpha - \langle A \rangle)^2 \, \rho(\alpha) \, d\alpha.$$

Continuous spectra and uncertainty relations

For a continuous spectrum with probability density $\rho(\alpha)$, the dispersion is given by:

$$(\Delta A)^2 = \int_{-\infty}^{+\infty} (\alpha - \langle A \rangle)^2 \, \rho(\alpha) \, d\alpha.$$

Equivalently:

$$(\Delta A)^2 = \int \alpha^2 \rho(\alpha) \, d\alpha - \left(\int \alpha \rho(\alpha) \, d\alpha \right)^2.$$

Continuous spectra and uncertainty relations

For a continuous spectrum with probability density $\rho(\alpha)$, the dispersion is given by:

$$(\Delta A)^2 = \int_{-\infty}^{+\infty} (\alpha - \langle A \rangle)^2 \, \rho(\alpha) \, d\alpha.$$

Equivalently:

$$(\Delta A)^2 = \int \alpha^2 \rho(\alpha) \, d\alpha - \left(\int \alpha \rho(\alpha) \, d\alpha \right)^2.$$

Applying this definition to non-commuting observables leads to the Heisenberg uncertainty relations:

$$\Delta X \, \Delta P_x \geq \frac{\hbar}{2}.$$

Continuous spectra and uncertainty relations

For a continuous spectrum with probability density $\rho(\alpha)$, the dispersion is given by:

$$(\Delta A)^2 = \int_{-\infty}^{+\infty} (\alpha - \langle A \rangle)^2 \, \rho(\alpha) \, d\alpha.$$

Equivalently:

$$(\Delta A)^2 = \int \alpha^2 \rho(\alpha) \, d\alpha - \left(\int \alpha \rho(\alpha) \, d\alpha \right)^2.$$

Applying this definition to non-commuting observables leads to the Heisenberg uncertainty relations:

$$\Delta X \, \Delta P_x \geq \frac{\hbar}{2}.$$

Compatible observables

Consider two observables A and B such that

$$[A, B] = 0.$$

Compatible observables

Consider two observables A and B such that

$$[A, B] = 0.$$

If both spectra are discrete, there exists a common basis of simultaneous eigenvectors:

$$A\,|a_n, b_p, i\rangle = a_n\,|a_n, b_p, i\rangle\,,$$
$$B\,|a_n, b_p, i\rangle = b_p\,|a_n, b_p, i\rangle\,.$$

Compatible observables

Consider two observables A and B such that

$$[A, B] = 0.$$

If both spectra are discrete, there exists a common basis of simultaneous eigenvectors:

$$A\,|a_n, b_p, i\rangle = a_n\,|a_n, b_p, i\rangle\,,$$
$$B\,|a_n, b_p, i\rangle = b_p\,|a_n, b_p, i\rangle\,.$$

Thus A and B admit a **common eigenbasis**.

Expansion of the initial state

An arbitrary normalized state can be written as:

$$|\psi\rangle = \sum_{n,p,i} c_{npi} \, |a_n, b_p, i\rangle \, .$$

Expansion of the initial state

An arbitrary normalized state can be written as:

$$|\psi\rangle = \sum_{n,p,i} c_{npi} \, |a_n, b_p, i\rangle \, .$$

The coefficients satisfy:

$$\sum_{n,p,i} |c_{npi}|^2 = 1.$$

Measure A then B

Probability of obtaining a_n:

$$\mathcal{P}(a_n) = \sum_{p,i} |c_{npi}|^2.$$

Measure A then B

Probability of obtaining a_n:

$$\mathcal{P}(a_n) = \sum_{p,i} |c_{npi}|^2.$$

After obtaining a_n, the state becomes:

$$|\psi_{a_n}\rangle = \frac{1}{\sqrt{\sum_{p,i} |c_{npi}|^2}} \sum_{p,i} c_{npi} \, |a_n, b_p, i\rangle \, .$$

Measure A then B

Probability of obtaining a_n:

$$\mathcal{P}(a_n) = \sum_{p,i} |c_{npi}|^2.$$

After obtaining a_n, the state becomes:

$$|\psi_{a_n}\rangle = \frac{1}{\sqrt{\sum_{p,i} |c_{npi}|^2}} \sum_{p,i} c_{npi} |a_n, b_p, i\rangle .$$

Conditional probability of obtaining b_p:

$$\mathcal{P}_{a_n}(b_p) = \frac{\sum_i |c_{npi}|^2}{\sum_{p,i} |c_{npi}|^2}.$$

Joint probability

The joint probability of obtaining a_n then b_p is:

$$\mathcal{P}(a_n, b_p) = \mathcal{P}(a_n)\,\mathcal{P}_{a_n}(b_p).$$

Joint probability

The joint probability of obtaining a_n then b_p is:

$$\mathcal{P}(a_n, b_p) = \mathcal{P}(a_n)\,\mathcal{P}_{a_n}(b_p).$$

Substituting:

$$\boxed{\mathcal{P}(a_n, b_p) = \sum_i |c_{npi}|^2.}$$

Measure B then A

Similarly,

$$\mathcal{P}(b_p, a_n) = \sum_i |c_{npi}|^2.$$

Measure B then A

Similarly,

$$\mathcal{P}(b_p, a_n) = \sum_i |c_{npi}|^2.$$

Therefore:

$$\boxed{\mathcal{P}(a_n, b_p) = \mathcal{P}(b_p, a_n).}$$

Measure B then A

Similarly,

$$\mathcal{P}(b_p, a_n) = \sum_i |c_{npi}|^2.$$

Therefore:

$$\boxed{\mathcal{P}(a_n, b_p) = \mathcal{P}(b_p, a_n).}$$

For compatible observables, the order of measurement does not matter.

Physical interpretation (compatible case)

If A and B commute:

Physical interpretation (compatible case)

If A and B commute:

- The order of measurements is irrelevant.
- No information is lost.
- The final state is a common eigenvector $|a_n, b_p, i\rangle$.

Physical interpretation (compatible case)

If A and B commute:

- The order of measurements is irrelevant.
- No information is lost.
- The final state is a common eigenvector $|a_n, b_p, i\rangle$.

Compatible observables can be measured simultaneously.

Incompatible observables

If

$$[A, B] \neq 0,$$

Incompatible observables

If

$$[A, B] \neq 0,$$

there is no complete set of common eigenvectors.

Incompatible observables

If

$$[A, B] \neq 0,$$

there is no complete set of common eigenvectors.
In general, a state cannot be a simultaneous eigenvector of both observables.

Two-dimensional illustration

Consider a two-dimensional Hilbert space.

Two-dimensional illustration

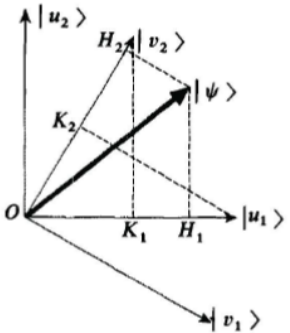
Consider a two-dimensional Hilbert space.
$\{|u_1\rangle, |u_2\rangle\}$ eigenbasis of A,
$\{|v_1\rangle, |v_2\rangle\}$ eigenbasis of B.

Two-dimensional illustration

Consider a two-dimensional Hilbert space.
$\{|u_1\rangle, |u_2\rangle\}$ eigenbasis of A,
$\{|v_1\rangle, |v_2\rangle\}$ eigenbasis of B.
Since A and B do not commute, the two bases do not coincide.

Two-dimensional illustration

Consider a two-dimensional Hilbert space.
$\{|u_1\rangle, |u_2\rangle\}$ eigenbasis of A,
$\{|v_1\rangle, |v_2\rangle\}$ eigenbasis of B.
Since A and B do not commute, the two bases do not coincide.



Measurement sequence (non-commuting case)

First measure A and obtain a_1:

$$|\psi\rangle \to |u_1\rangle \,.$$

Measurement sequence (non-commuting case)

First measure A and obtain a_1:

$$|\psi\rangle \to |u_1\rangle \,.$$

Then measure B and obtain b_2:

$$|u_1\rangle \to |v_2\rangle \,.$$

Measurement sequence (non-commuting case)

First measure A and obtain a_1:

$$|\psi\rangle \to |u_1\rangle \,.$$

Then measure B and obtain b_2:

$$|u_1\rangle \to |v_2\rangle \,.$$

Reverse order:

$$|\psi\rangle \to |v_2\rangle \to |u_1\rangle \,.$$

Order matters when $[A, B] \neq 0$

In general:

$$\mathcal{P}(a_1, b_2) = |OH_1|^2 \, |OK_2|^2,$$

Order matters when $[A, B] \neq 0$

In general:

$$\mathcal{P}(a_1, b_2) = |OH_1|^2 \, |OK_2|^2,$$

$$\mathcal{P}(b_2, a_1) = |OH_2|^2 \, |OK_1|^2.$$

Order matters when $[A, B] \neq 0$

In general:

$$\mathcal{P}(a_1, b_2) = |OH_1|^2 \, |OK_2|^2,$$

$$\mathcal{P}(b_2, a_1) = |OH_2|^2 \, |OK_1|^2.$$

Although $|OK_1| = |OK_2|$, in general $|OH_1| \neq |OH_2|$.

Order matters when $[A, B] \neq 0$

In general:

$$\mathcal{P}(a_1, b_2) = |OH_1|^2 \, |OK_2|^2,$$

$$\mathcal{P}(b_2, a_1) = |OH_2|^2 \, |OK_1|^2.$$

Although $|OK_1| = |OK_2|$, in general $|OH_1| \neq |OH_2|$.

$$\boxed{\mathcal{P}(a_1, b_2) \neq \mathcal{P}(b_2, a_1)}$$

Loss of information

When A and B do not commute:

Loss of information

When A and B do not commute:

- The second measurement disturbs the system.
- Information obtained in the first measurement can be lost.
- The final state depends on the order.

Loss of information

When A and B do not commute:

- The second measurement disturbs the system.
- Information obtained in the first measurement can be lost.
- The final state depends on the order.

Incompatible observables cannot be measured simultaneously.

Preparation of a state

If measuring A yields a non-degenerate eigenvalue a_n, the system collapses to:

$$\left|u_n\right\rangle .$$

Preparation of a state

If measuring A yields a non-degenerate eigenvalue a_n, the system collapses to:

$$|u_n\rangle .$$

The final state does not depend on the initial state, only on the result obtained.

Preparation of a state

If measuring A yields a non-degenerate eigenvalue a_n, the system collapses to:

$$|u_n\rangle .$$

The final state does not depend on the initial state, only on the result obtained.

However, if the eigenvalue is degenerate, the post-measurement state depends on the initial coefficients.

Complete set of commuting observables

To uniquely determine the post-measurement state, one must measure a complete set of commuting observables (C.S.C.O.).

Complete set of commuting observables

To uniquely determine the post-measurement state, one must measure a complete set of commuting observables (C.S.C.O.).

A C.S.C.O. allows us to prepare a unique basis state of the Hilbert space.

Complete set of commuting observables

To uniquely determine the post-measurement state, one must measure a complete set of commuting observables (C.S.C.O.).

A C.S.C.O. allows us to prepare a unique basis state of the Hilbert space.

Changing the set of observables allows us to prepare different states.